

Weekly Report – The Hybrid Model

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This Week

- **Two U-Nets compared:** Akeret et al.'s `tf_unet` (ours) vs. the U-Net of Mesarcik et al.
- **Hybrid model:** presented for the first time — a U-Net with 9 changes.
- **Which changes matter:** tested by removing parts and re-training.

Key Results (LOFAR, max F1)

- `tf_unet`: 0.548
- Mesarcik et al. U-Net: 0.588 (+0.039)
- Our hybrid: **0.660** — highest we found (best published: 0.640)
- Hybrid with changes 1–3 removed: 0.658 — no real loss

➤ Two different U-Nets

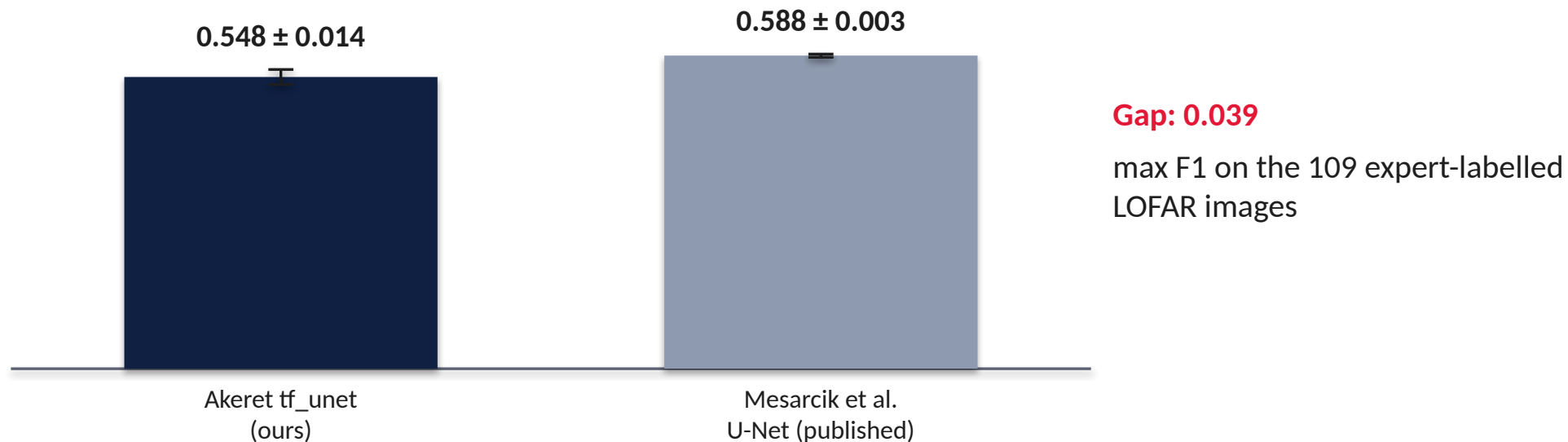
Akeret et al. (2017) — tf_unet

The original U-Net code for RFI detection, written in TensorFlow.

This is the U-Net we ran.

Mesarcik et al. (2022) — their own U-Net

Rebuilt from scratch by the authors of the LOFAR dataset paper. Same name, but a different network (see next slide).



➤ What is different between them

	Akeret et al. tf_unet (ours)	Mesarcik et al. U-Net
Normalisation layers	None	BatchNorm after every layer
Shrinking the image	Max-pooling, 2 times (3 levels)	Stride-2 convolutions, 5 times
Image edges	Trimmed (valid padding)	Kept (same padding)
Network size	465,986 parameters	1,179,121 parameters
Training input	Whole 512×512 image, 4 per batch	32×32 patches, 1024 per batch
Input scaling	Fixed linear range	Clipped, then log-scaled
Output	2 scores, ReLU, softmax	1 score, sigmoid

So the 0.039 gap compares two different networks — not the same U-Net trained twice.

➤The 9 changes in our Hybrid model (1 of 3)

1

Strip convolutions

- A normal filter looks at a small 3×3 square of pixels.
- Each strip path chains a $1 \times K$ and a $K \times 1$ filter ($K = 7, 11, 21$): together they cover a $K \times K$ square.
- Idea: RFI appears as long streaks. As built, it looks at a wider area, not a thin line.

2

Residual shortcuts

- Normally the data must pass through every layer, one after another.
- A shortcut lets the input skip past a block and be added back at the end.
- Idea: faint details are not lost as the data goes deeper into the network.

3

Channel attention (ECA)

- The network builds many feature maps, each looking for a different pattern.
- Attention learns a weight for each one: useful ones up, others down.
- Idea: help the features that spot faint RFI stand out.

➤The 9 changes in our Hybrid model (2 of 3)

4

GroupNorm

- Inside a network, numbers can grow too large or shrink too small.
- GroupNorm rescales them inside every block so they stay in a steady range.
- tf_unet has no such step; this also makes training more stable.

5

No ReLU on the output

- tf_unet passes its final scores through a ReLU, which turns negatives into 0.
- With class weighting this can freeze the network so that it stops learning.
- The hybrid outputs its raw scores, so this trap cannot happen.

6

Dice loss

- The usual loss (cross-entropy) checks every pixel separately.
- Dice loss also checks how well the whole predicted RFI shape overlaps the true one.
- Helpful when RFI is rare: about 1 pixel in 130 on LOFAR.

➤The 9 changes in our Hybrid model (3 of 3)

7

Same padding

- tf_unet trims the border of the image at every layer (valid padding).
- Same padding adds a thin border so the image keeps its full size.
- So every pixel, including the edges, gets a prediction.

8

No class weighting

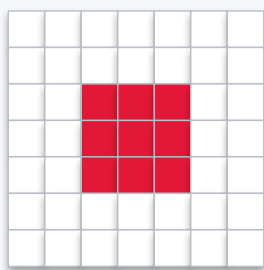
- Class weighting tells the model to care more about the rare RFI pixels.
- Our earlier version used it; removing it improved every run (0.647 → 0.660).
- tf_unet does not use it either, so this is not a difference from tf_unet.

9

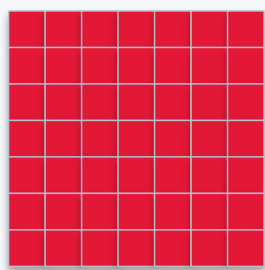
New size and training setup

- Shrinks the image 4 times instead of 2, but thinner: 8 filters in the first layer, not 32.
- Written in PyTorch; trained with Adam, one image at a time, 28,000 steps.
- About 594,000 parameters in total.

1. Strip convolutions



normal 3×3 filter

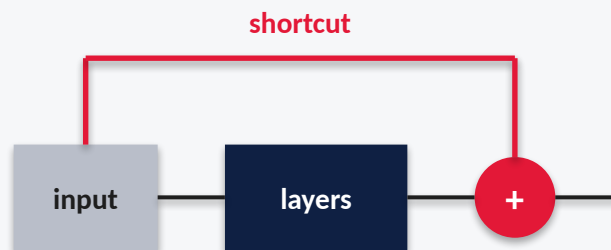


strip path: 1×7 then 7×1

Chains a 1×7 and a 7×1 filter (also 11 and 21). Together they cover a whole square, not a line.

Idea: catch streaks. As built, it sees a wider area.

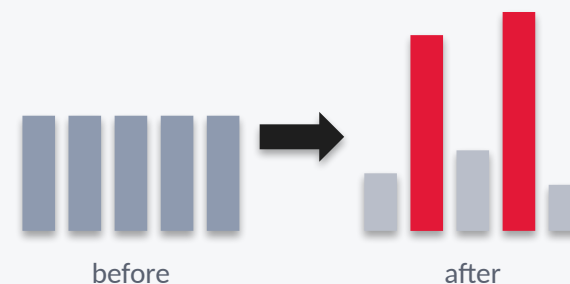
2. Residual shortcuts



A bypass road: the input skips past the block and is added back at the end.

Idea: faint details are not lost in deep layers.

3. Channel attention (ECA)



Learns which feature maps matter and turns them up or down.

Idea: let faint-RFI features stand out.

➤ Do changes 1-3 matter?

Full hybrid

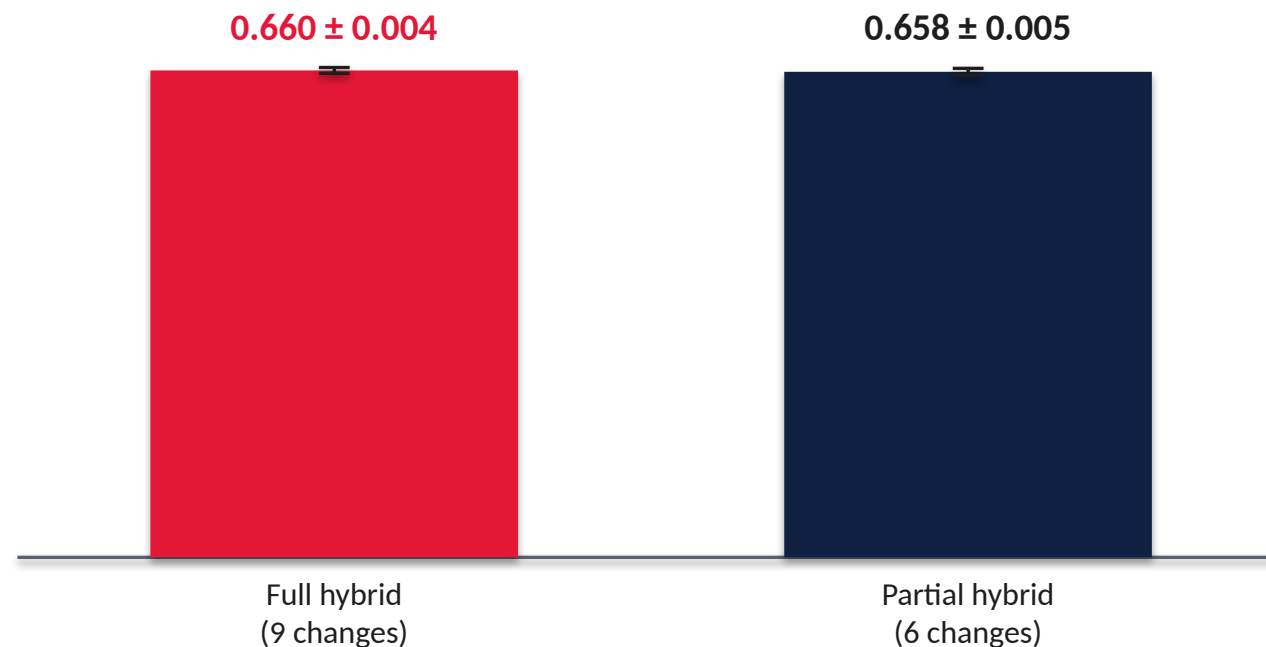
All 9 changes

593,842 parameters

Partial hybrid

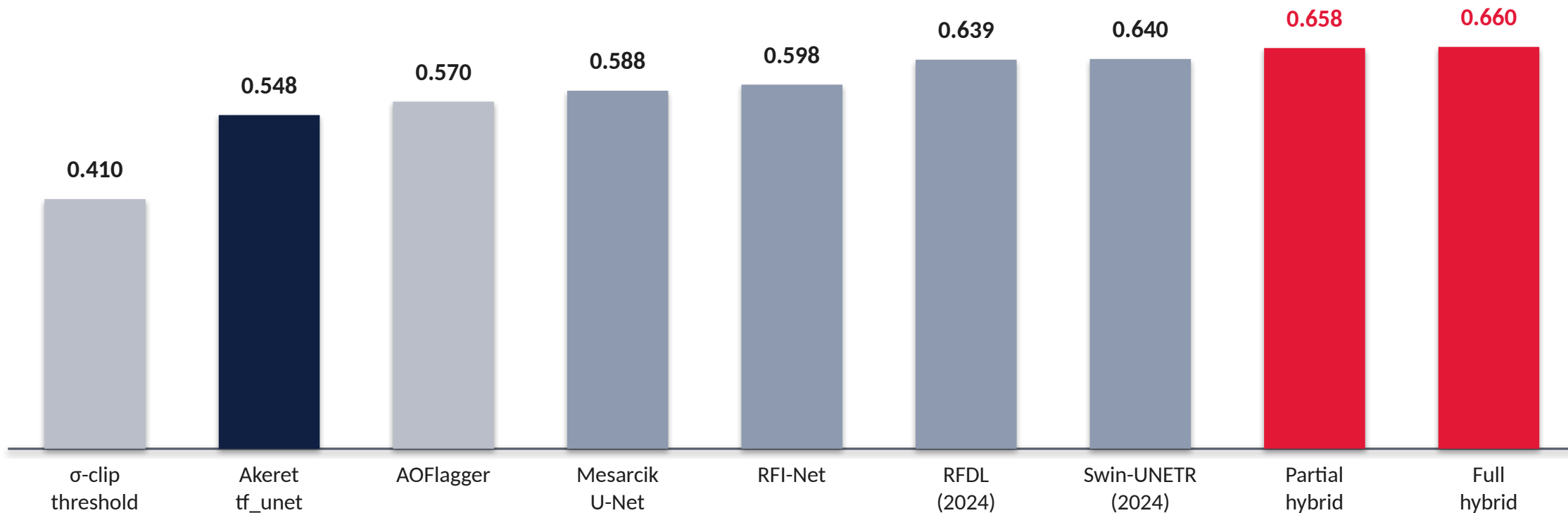
Changes 1-3 removed (no strips, no shortcuts, no attention): 6 changes left

486,418 parameters (18% smaller)



No — the score barely moves: 0.660 vs 0.658 (3 runs each, $p = 0.64$).
Strips, shortcuts and attention add 22% more parameters but no accuracy.

➤ Where everything stands on LOFAR



max F1 on the 109 expert-labelled LOFAR images; every network was trained on AOFlagger labels. Mesarcik U-Net, RFI-Net: Mesarcik et al. (2022); RFDL: van Zyl & Grobler (2024); Swin-UNETR (400M parameters): Ouyang et al. (2024).

Our lead over RFDL and Swin-UNETR (about 0.02) is smaller than the test-set uncertainty (± 0.04).

➤ To Do Next Week

- **Find what drives the gain:** test changes 5, 6, 7 and 9 one at a time, starting with the Dice loss.
- **Repeat runs** where results so far come from a single run.
- **Project report:** continue from the full first draft.